

St. Margaret's Co-educational English Secondary & Primary School
S.3 Mock Practice Paper, 2025–2026
S.3 Information and Communication Technology
Marking Scheme

Quiz — Part A: Matching (5 marks)

Section A: Matching — Python Code Snippets

Q1.	B. Performs subtraction and stores the result in a variable. <code>total = a - b</code>
Q2.	C. Captures user input and assigns it to a variable as a string. <code>ans = input("?")</code>
Q3.	A. Assigns a float (decimal) value to a variable. <code>price = 19.99</code>
Q4.	D. Outputs a literal string or variable to the console. <code>print("Hello")</code>
Q5.	E. Converts a value into an integer number. <code>int(n)</code>

Quiz — Part B: Short Questions (5 marks)

B1. (2 marks)	The machine classifies the image as Carrot Cake. Its confidence value (0.97363) is closer to 1, indicating a much higher probability. A value near 1 means higher classification confidence; a value near 0 means lower confidence. Siu Mai's value (0.02603) is near 0, so it is very unlikely to be the classification.
B2. (2 marks)	Any two of: (1) Increase the total number of images — at least 10 per label. (2) Capture images from different angles. (3) Include diverse variations of the object. (4) Ensure consistent and improved lighting. (1 mark each.)
B3. (1 mark)	TextToSpeech

Written Exam — Part A: Multiple-Choice Questions (10 marks, 1 mark each)

Q1.	C. <code>_total</code> Python variable names can start with a letter or underscore, not a digit, hyphen, or space.
Q2.	B. <code>TypeError</code> Python cannot concatenate a <code>str</code> with an <code>int</code> using <code>+</code> ; type conversion is required.
Q3.	D. <code>String (str)</code> <code>input()</code> always returns a string regardless of what the user types.
Q4.	C. <code>Knowledge-based Reasoning</code> Uses in-depth analysis using prior experience, process knowledge, and objectives — the highest level of AI reasoning.
Q5.	C. <code>Data Entry Clerks</code> Data entry is highly automatable. AI/ML Specialists, Big Data Specialists, and IoT Specialists are all growing roles.
Q6.	B. "Immutable" means individual characters in a string cannot be modified directly once

	created; the variable can be reassigned, but the string object itself cannot be altered in place.
Q7.	B. WebViewer WebViewer displays the real-time camera feed so users can aim their phone at objects for classification.
Q8.	C. The Global Digital Divide Refers to the gap between those with ICT access and those without.
Q9.	B. Supervised learning uses paired input–output training data provided by a supervisor to train the model.
Q10.	C. Repetitive Strain Injury (RSI) RSI results from damage to nerves and tendons caused by repetitive movements; Carpal Tunnel Syndrome is a specific type of RSI.

Written Exam — Part B: Matching (10 marks, 1 mark each)

1.	B. Assigns an integer value to a variable. <code>score = 100</code>
2.	D. Assigns a string (text) value to a variable. <code>city = "London"</code>
3.	C. Checks the data type of a variable. <code>type(x)</code>
4.	E. A conditional check to see if x is not equal to 5. <code>if not x == 5:</code>
5.	A. A conditional check for less than or equal to. <code>if x <= y:</code>
6.	G. Assigns a float value to a variable. <code>price = 19.99</code>
7.	F. Performs subtraction and stores the result. <code>total = a - b</code>
8.	H. Outputs a literal string to the console. <code>print("Hello")</code>
9.	I. Captures user input as a string. <code>ans = input("?")</code>
10.	J. Converts a value into an integer. <code>int(n)</code>

Written Exam — Part C: Fill in the Blanks (10 marks, 1 mark each)

C1.	Training Accept: "model training"
C2.	PersonalImageClassifier.aix Must include the .aix extension for full mark.
C3.	Supervised Learning / Reinforcement Learning Both blanks required (0.5 per blank).
C4.	Knowledge Accept: "Knowledge-based"

C5.	Indentation Accept any correct spelling variation.
C6.	Text Neck Both words required.
C7.	TypeError Accept: "Type Error"
C8.	Cognitive and Metacognitive Both words required.
C9.	background Accept: "socioeconomic background", "financial background", "social background"
C10.	letter Accept: "a letter" / "alphabet letter"

Written Exam — Part D: Questions (20 marks)

D1.	App Development with MIT App Inventor (6 marks)
(a) (3 marks)	Award ½ mark per step, up to 3 marks (6 steps total): 1. Access the Classifier Tool (https://classifier.appinventor.mit.edu/). 2. Create Labels (categories). 3. Add Training Data — at least 10 images per label. 4. Training — allow the model to learn from images. 5. Testing — provide additional images to test accuracy. 6. Download the Trained Model.
(b) (2 marks)	1 mark each. Any two of: WebViewer / TextToSpeech / Camera / PersonalImageClassifier (the imported .aix extension).
(c) (1 mark)	Reason: The model lacks diversity in training data — all images were taken from the same angle and under identical lighting. Fix: Capture images from different angles and under varied lighting conditions.
D2.	Python Basics (8 marks)
(a) (4 marks)	Error 1 (Lines 1–2) — 2 marks: <code>input()</code> always returns a string; multiplying by 3 causes a <code>TypeError</code> . Fix: <code>A = int(input())</code> — convert input to an integer first. Error 2 (Line 3) — 2 marks: Cannot concatenate a string with integer B using <code>+</code> . Fix: <code>print("The answer is: " + str(B))</code> or <code>print("The answer is:", B)</code> .
(b) (3 marks)	1 mark per correct line. Accept any valid equivalent: <code>name = input("Enter your name: ")</code> <code>num = input("Enter your class number: ")</code> <code>print("Hello " + name + ", you are in class S" + num + ".")</code>

	Note: str() conversion not needed if num stays a string from input().
(c) (1 mark)	(i) bool — True and False are Boolean values. (ii) float — 3.0 has a decimal point, so it is a float (not an int).
D3.	AI and the Future of Work (6 marks)
(a) (2 marks)	Negative (1 mark): AI can automate many tasks, leading to job displacement (e.g., Data Entry Clerks). Positive (1 mark): AI also creates new job opportunities (e.g., AI/ML Specialists, Data Scientists) and increases efficiency across industries.
(b) (2 marks)	Knowledge-based Reasoning is most suitable. (1 mark) It uses in-depth analysis of the current situation, prior experience, and full knowledge of distances between locations to determine the shortest delivery path — yielding the most efficient route. (1 mark) Skill-based only makes direct roundtrips; Rule-based uses fixed patterns that may not apply to every situation.
(c) (2 marks)	1 mark per skill + explanation pair. Must be from DIFFERENT OECD categories. Example: Empathy (Social & Emotional Skills): AI cannot replicate genuine human understanding and care, so empathy allows workers to build relationships and support others in ways AI cannot. Creative Thinking (Cognitive & Metacognitive Skills): AI can process data but struggles with original ideas; creative thinking helps humans innovate and solve problems in new ways. Accept any valid skills from different categories with a reasonable explanation.

All questions may have more than one acceptable answer. Other reasonable answers should be considered.

End of Marking Scheme